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A Quality-Oriented Digital Twin Modelling Method for Manufacturing Processes Based on A Multi-Agent Architecture

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Abstract

The quality of a product is highly dependent on manufacturing processes. The recent development of industrial information technologies, such as Cyber-Physical Production Systems, Industrial Internet of Things, and Big Manufacturing Data Analytics has empowered the digitalization of manufacturing processes and promoted the concept of Digital Twin (DT). As one of the fundamental enabling technologies for Industry 4.0, DT enables the convergence between a physical system and its digital representation. DT modelling is the basis of implementing DT in practice. In this paper, we propose a DT modelling method based on a multi-agent architecture. It focuses on quality control during manufacturing processes and provides solutions to gather relevant information and analyze the corresponding influences on product quality. The MPFQ-model (Material, Production Process, Product Function/Future, Product Quality) is adopted to support the analysis of main influential factors related to the final product quality during the manufacturing phase. The five-dimension architecture is used as the basis for the DT models, including (i) physical entities, (ii) virtual models, (iii) DT data, (iv) services and (v) connections. Based on this architecture a Multi-Agent System (MAS) component and a semantic engineering component are integrated to create a quality-oriented DT framework.

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Keywords: Digital Twin; multi-agent system; quality control; manufacturing processes;

1. Introduction

The rapid advancement of technologies such as Internet of Things (IOT), Artificial Intelligence (AI), Cyber-Physical Production Systems (CPS) has reshaped different aspects of the modern society including the industry sector. Empowered by these technologies, new paradigms such as Cyber-Physical Production Systems (CPPS), Industrial Internet of Things (IIOT), and Big Manufacturing Data Analytics have been widely deployed in modern manufacturing enterprises [1–5]. These advanced paradigms have accelerated the digitalization of manufacturing processes and fostered the concept of Digital Twin (DT). As one of the fundamental enabling technologies for the Factories of the Future (FoF), DT enables the convergence between a physical system and its digital representation [6,7]. It has been widely used for modelling and simulation of system development to support design or to validate system properties [8,9], optimization of manufacturing processes and services, and prediction of failures and product defects [2,10] etc. DT modelling is the basis of implementing DT in practice. Due to the complexity of the physical systems and processes, it is critical to build a feasible DT model which can properly represent the actual conditions associated with a physical system and ensure a correct match between the original physical system and the constructed DT model [2].

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The quality of a product is highly dependent on production design, manufacturing processes, and the inbound logistic purchasing all materials needed for the manufacturing of the product [11]. All these factors converge during the manufacturing phase in a factory where the designed product quality is expected to be realized by machining and assembling procured materials according to proper production processes [11]. Therefore, the interactions among these factors need to be considered when designing a DT model for a manufacturing shop-floor.

The aim of this paper is to propose a DT modelling method taking into account the main elements related to quality, including material, production process, product functions, product quality. It focuses on quality control during manufacturing processes and provides solutions to gather relevant information and analyze the corresponding influences on product quality. A component based on Multi-Agent Systems (MAS) is included in the model to collect information related to different aspects through autonomous and cooperative agents. A knowledge management module based on ontology is introduced to support the interactions among different agents.

The rest of the paper is arranged as follows: Section 2 introduces some existing studies related to this paper. Section 3 presents the architecture of the proposed DT modelling method including the explanation to each component. Section 4 introduces the main enabling technologies used in the modelling method. Section 5 summarizes the paper with a conclusion about the preliminary findings and contributions of the proposed approach and the outlook into the future work.

2. Related work

Modern manufacturing enterprises are facing challenges to build flexible and adaptive manufacturing systems to produce the right quantity of products with proper quality which is especially enforced by a higher accurateness of the quality attributes [12]. Engineering of production lines suffers a big pressure to become more flexible, efficient and to reduce costs [13]. To deal with the complex quality requirements, the EU-project “inteGration of pRocess and quAlity Control using multi-agEnt technologies” (GRACE) was created. The main focus of the GRACE project is set on the provision of a new Multi-Agent Architecture, combined with a methodology for quality assurance and improvement to provide suitable solutions for plant operators [11]. The MPFQ-model was developed in this project to provide integrated consideration of product quality covering material, production process, product functions. MAS is considered as a suitable technology in the development of the concept of FoF, addressing the requirements of adaptation, robustness and responsiveness [14]. The MAS principles were used in GRACE project to design distributed control systems based on autonomous and cooperative agents that interact with each other for the realization of efficient, flexible and robust systems.

In recent years, the concept of DT is getting increasingly popular both in academia and industrial applications. DT is not a totally new concept. It was fostered by the various existing technologies such as IIoT, CPPS, AI, 3D modeling, system simulation, digital prototyping etc. [15,16] In the whitepaper [6] published in 2014, Grieves defined the concept of DT and proposed a three-dimension model of DT based on the previous conception of “a virtual, digital equivalent to a physical product”. According to Grieves, a DT model should at least consist of three main parts including: physical products in Real Space; virtual products in Virtual Space; and the connections of data and information that tie the virtual and real products together [6]. In recent years, DT and relevant technologies have been evolving rapidly, which reflects that the virtual world and the physical world are becoming increasingly linked to each other and integrated. Based on the existing three-dimension DT model the authors of [7] proposed a five-dimension model to promote the further applications of DT by adding two more dimensions, i.e. DT data and services. A recent study [16] reviewed the application fields, enabling technologies and tools for DT. Based on this study, it is concluded that universal design and development platforms and tools for DT are required to facilitate the integration of different technologies and tools which may have different formats, protocols and standards.

DT has been widely used in manufacturing for optimized operations and failure prediction. An on-board integrated vehicle health management system was developed using DT as a paradigm shift to enable flight safety and reliability [17]. A DT reference model for rotating machinery fault diagnosis was designed which could bring advanced computing capability into machinery health management [2]. The authors of [18] established a five-dimension DT model consisting of (i) physical entities, (ii) virtual models, (iii) DT

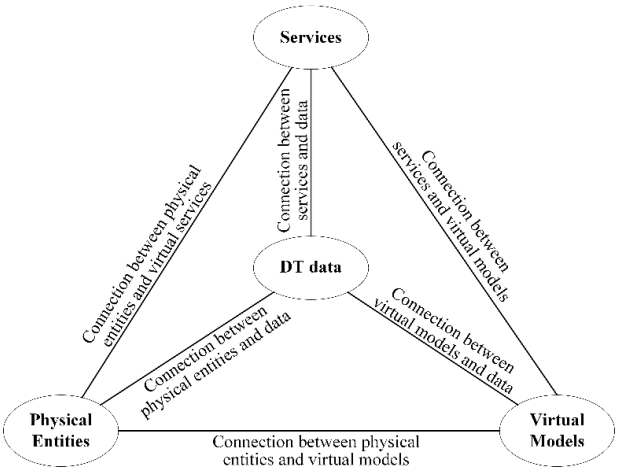


Figure 1 Five-dimension digital twin model

data, (iv) services and (v) connections, as shown in Figure 1, for the prognostics and health management of complex equipment.

To handle the heterogeneous data from different platforms and sources, semantic technologies have been employed in some recent studies [19–21]. Semantic models can capture complex systems in an intuitive fashion, which

can be summarized in standardized ontology languages, and come with a wide range of off-the-shelf systems to design, maintain, query, and navigate semantic models [22]. This characteristic makes semantic modelling a promising paradigm to address the challenges that DT development is facing currently. In a recent study [23], by integrating augmented semantic capabilities with DT, the concept of Cognitive Twin (CT) was proposed for identifying the dynamics of virtual model evolution, and enhancing the decision-making based on DT.

3. Architecture

Based on the existing studies of DT modelling and the output of the previous GRACE project, in this study we propose a quality-oriented DT modelling method for manufacturing processes by integrating traditional DT architectures with the MPFQ-model and ontology-based semantic knowledge management. The general architecture of the proposed DT modelling method is presented in Figure 1. It is composed of five components, including the physical production system, virtual entities of the physical system, a MAS component, an ontology-based semantic knowledge management layer and a service layer. The details and functions of each component are introduced as follows.

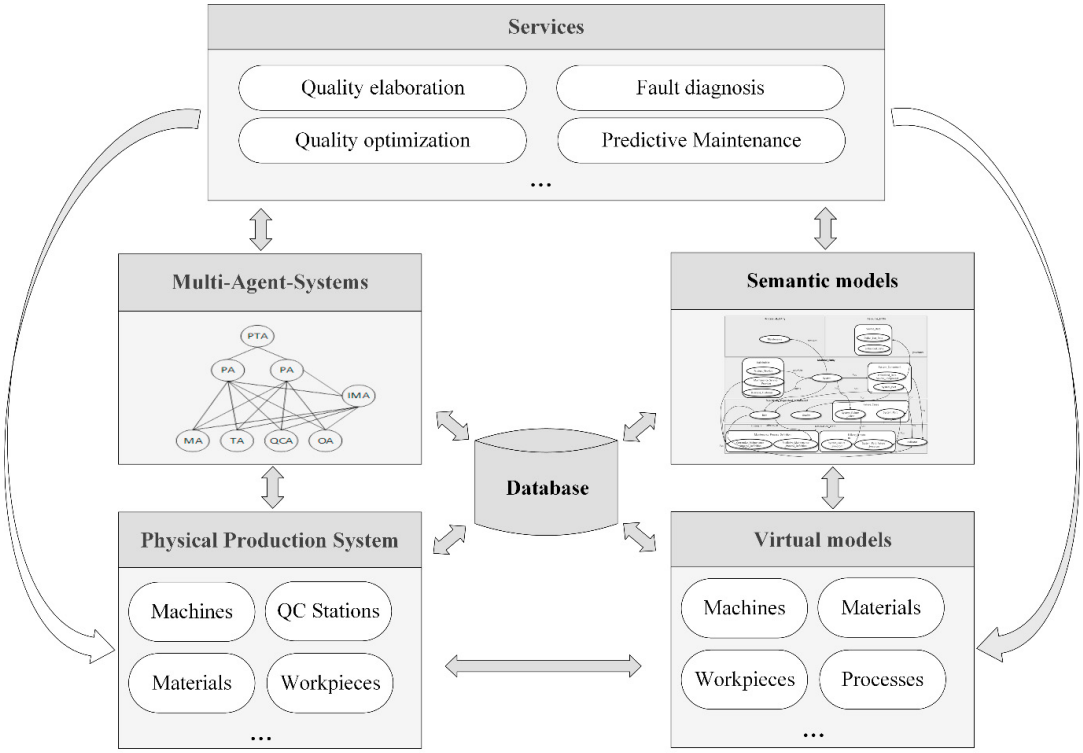


Figure 2 The architecture of the DT modelling method

- **Physical entities**: The basic purpose of DT is to create the virtual models in the virtual space for physical entities in real space, aiming to simulate their behaviors [6,16]. The physical entities may consist of device or product, physical system, activities process, even an organization. According to their functions and structures, the physical entities can be divided into three levels, i.e. unit level, system level, and system of system (SoS) level [24]. In this study, we focus mainly on the system level corresponding to the physical production system which may be consist of product instances, materials and the physical resources of the production line, such as machines, robots, quality control stations etc. The sensory devices contained in the physical systems can collect various data reflecting the conditions of the production system. They are also the foundation for creating the digital models in the virtual space.

- **Virtual entities**: They are faithful replicas of physical entities, which integrate digital engineering models and multiple variables, scales and abilities of the physical entities to reproduce its geometries, physical properties, behaviors and rules in the virtual space [7,18]. The geometric models describe the shape, size, tolerance, and structural relation of the physical entity; the property models contain the physical phenomena like the deformation, delamination, fracture and corrosion of the physical entity; the behavior models define the behaviors, such as state transition, performance degradation and coordination, and responding mechanisms of the physical entity against changes in the external environment [16]. These different type of models in the virtual space are closely related with each other, i.e. the behavior models are defined on the basis of the property models, which themselves are based on the geometry models. The

functions of the behavior models also require the support of predefined rule models. In some studies [7,16,18], the rule models are defined as part of the virtual entities. In this study, we separated this function from the virtual entities and introduced two additional components, i.e. MAS and ontology, to realize the functions of reasoning, judgement, evaluation, and autonomous decision-making etc.

- **MAS component**: In this study we reused the MAS architecture developed in the GRACE project, which comprises a series of autonomous and cooperative agents representing different manufacturing components disposed along a production system [11]. More details of the MAS architecture are introduced in the following sections about enabling technologies.

- **Semantic models**: The semantic models provide support to both the MAS and virtual entities' behavioral models. For the MAS agents, their interaction requires a proper knowledge representation to establish a common understanding among different agents. The ontologies define the vocabulary and the semantics of the knowledge used in the communication between distributed agents. For the virtual entities' behavioral models, the ontologies enable their logical capabilities, like reasoning and autonomous decision-making, by formalizing the structure of the knowledge such as the entities, the relations between entities and the attributes of each entity etc. The development of the ontology was based on the previously developpe MPFQ-model. More details are introduced in the following sections about enabling technologies.

- **Services**: This component provides feedbacks to both physical and virtual entities. On one hand, it enables application services such as production process monitoring and optimization, fault diagnosis and prognosis, equipment prognostic and health management (PHM), predictive maintenance etc. On the other hand, it ensures the high fidelity of the virtual entities through calibrating the parameters of the virtual models during the running to sustain its high performance with the physical entities.

4. Enabling technologies

4.1. MPFQ-model

In this study we used the MPFQ-model, which was developed in the previous GRACE project, to describe the relevant impact factors to product quality during the manufacturing phase while considering the strong interactions between product design and plant planning. As the name indicated, the MPFQ-model contains four main elements: Material (M), Production Process (P), Product Functions/Features (F) and Product Quality (Q). The definition of every element is explained as follows:

- Material represents everything that is needed to produce a certain product or product component, which may include raw materials, pre-products, consumables, operating supplies, product components and assemblies [25].

- Production Processes are defined as processing and transforming materials into the final goods by using machines, tools and human labor. This process is defined within the plant engineering [26].

- Product Functions / Features represent the distinguished characteristics of a product item, which may include functionalities like specific tasks, actions or processes that the product is able to perform; and/or other features like performance [11].

- Product Quality (Q) is defined as, according to DIN EN ISO 9000, the degree of conformance of final product functions and features to customer requirements.

The four elements of the MPFQ-model and the interrelations among them are illustrated in Figure 3 [11].

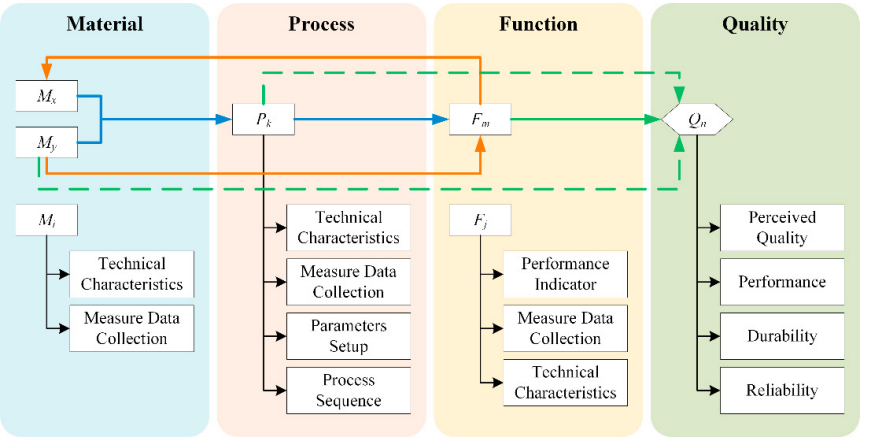


Figure 3 MPFQ-model elements and their interrelations

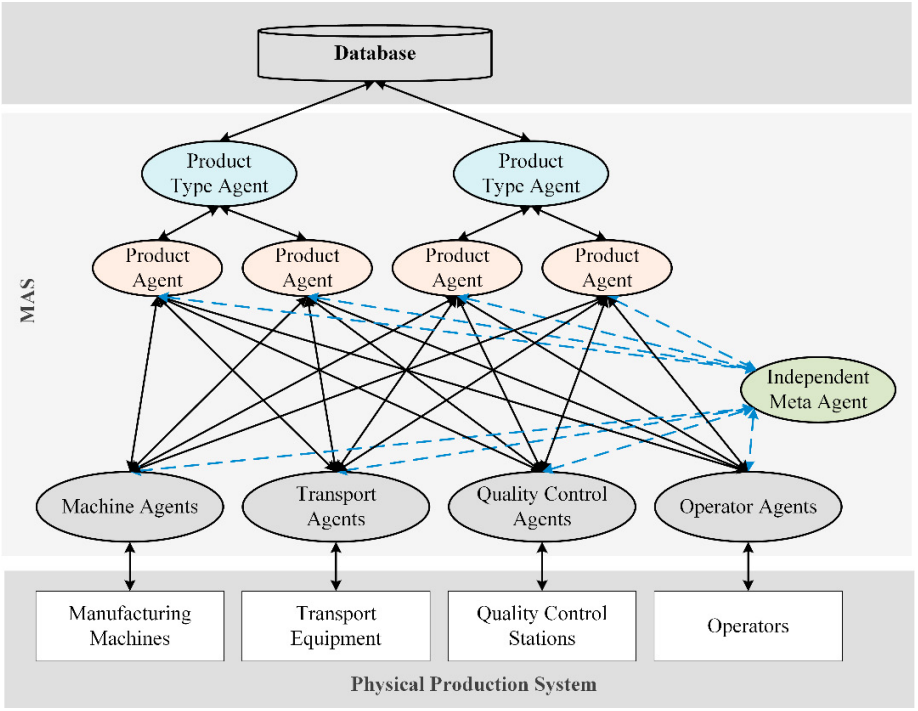


Figure 4 MAS structure and interactions among agents

4.2. MAS

The MAS is adopted in this study to cope with different manufacturing components of the production system. Following the process to control and to their specialization [27], a series of autonomous and cooperative agents were defined, including Product Type Agents (PTA), Product Agents (PA), Resource Agents (RA) and Independent Meta Agents (IMA). PTA represent the catalogue of products/parts that can be produced by the production line; PA handle the production of product instances along the production line; RA represent the physical resources of the production line, which can be divided into several sub-agents such as Machine Agents (MA), Quality Control Agents (QCA), Transport Agents (TA) and Operator Agents (OA); IMA implement global supervisory control and optimized planning and decision-making mechanisms. More details about the definitions and roles of these agents were introduce in the previous study [11]. The MAS structure and interactions among agents is demonstrated in Figure 4.

4.3. Ontology and semantic modelling

Data from different platforms and sources might be heterogeneous in syntax, schema, or semantics, which make data integration and data interoperability difficult. Ontology engineering and semantic modelling provide solutions to achieve semantic interoperability in a heterogeneous information system. Ontologies provide formal models of domain knowledge exploited in different ways. In some previous projects, some domain ontologies related to advanced manufacturing have been developed. These relevant ontologies can be used as references for the development of new ontologies.

The MPFQ-model provides fundamental elements related to product quality during the manufacturing process. These elements, i.e. materials, processes, functions, quality and the correlations among them, should all be built into the ontology. It will enable the interactions between different agents of the MAS component. An agent will be able to comprehend and interpret all data related to different elements of the MPFQ-model from other agents.

Creation of new entities of the ontology should start from the upper-level ontology like the Basic Formal Ontology (BFO). It supports information retrieval, analysis and integration; and facilitates interoperability among domain ontologies that are built in its terms through a process of downward population. The ontology development should follow the general architecture of Industrial Ontologies Foundry (IOF) (https://www.industrialontologies.org), which aims to co-create a set of open ontologies to support the manufacturing and engineering industry needs and advance data interoperability.

The ontology schema can be edited and validated with the open source ontology editor Protégé. It supports Web Ontology Language (OWL) and includes deductive classifiers to validate that models. It has an easy connection with agent development frameworks such as Java Agent Development Framework (JADE).

5. Implementation of the MAS-based modelling method

This section describes how the proposed MAS-based digital twin modelling method can be implemented to improve quality control based on a case of a discrete assembly line for washing machine. Figure 5 presents a MPFQ modelling example for the "seal water" quality of a washing machine. This example is derived from the overall MPFQ-model for a washing machine developed during the GRACE project [11]. As shown in Figure 5, the "seal water" quality is achieved by two functions "push" and "hold". The "push" function is realized by the "seal" and "shaft" materials which are connected by the "drum insertion" process. Similarly, the "hold" function is realized by the materials "seal" and "hub" which are connected by the

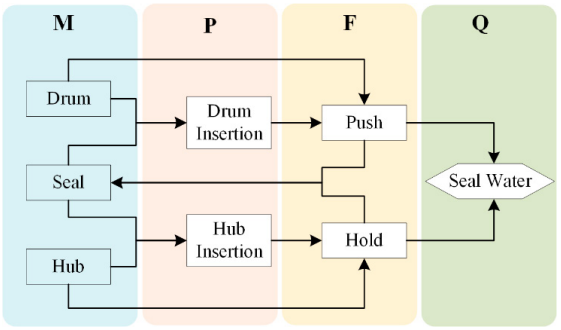


Figure 5 MPFQ-model for the "seal water" quality of a washing machine

"insertion" process.

During the production phase, different MAS agents will collect relevant data from the running production line. For example, MAS agents can measure the depth and force of the insertion during a drum insertion process; QCA agents can collect data directly from the product using their quality control stations. After the collection of raw data, MAS agents may also perform data elaboration and integration to generate further information regarding the quality of the product.

All the data including those collected directly and elaborated by the MAS agents will be stored in a nonstructural database which supports data retrieval following the semantic modelling standards. In this database, the data generated by the virtual models may also be saved. On the other hand, the data provided by the MAS agents will be used to update the virtual models. Empowered by the semantic models, these data will be the basis of many quality-oriented services such as quality elaboration, quality optimization, fault diagnosis, predictive maintenance and so on. Certain advanced data analytic tools, such as machine learning and deep learning, can be employed to support the data analysis. These services will in return provide feedbacks to the physical systems and virtual models.

6. Conclusions

This paper proposed a novel DT modelling method by integrating a MAS component and semantic models with the five-dimension DT model. This method aims to improve the quality control during the manufacturing phase considering all the factors that impact the quality of a product based on the MPFQ-model. The autonomous and cooperative agents of the MAS facilitate the interactions among different manufacturing components of a production line. The semantic models enable the common understanding during the interactions among different agents. These technologies make possible of quality-related services such as quality monitoring and optimization, fault diagnosis, predictive maintenance etc.

The presented modelling method will be implemented in the QU4LITY project (https://qu4lity-project.eu/), which is the biggest European project dedicated to Autonomous Quality (AQ) and Zero-Defect Manufacturing (ZDM) in the Industry 4.0. As following steps, more efforts will be spent on the development of domain ontology focusing on the manufacturing phase and quality related factors based on the MPFQ-model. A case study will be conducted to verify the feasibility of the proposed method.

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